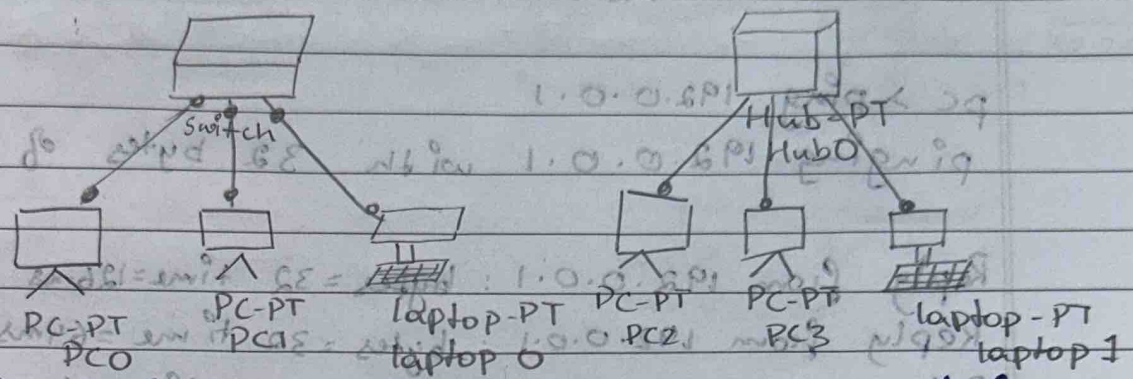
1) Create a topology & simulate sending a simple PDU from source to destination using hub and switch as connecting devices & demonstrate ping message.

Sol:- Step 1 : Create a simple topology like this.



Step 2: Configure IP address to each device as host as shown, click on device

Step 3: Click the IP configuration. Assign the class of IP address (class A, class B, class C)

Step 4: Repeat the same process to assign IP address for all hosts in topology

Host	IP address	Subnet Mask
PC0	192.0.0.1	255.255.255.0
PC1	192.0.0.2	255.255.255.0
laptop 0	192.0.0.3	255.255.255.0
PC 2	192.0.0.4	255.255.255.0
laptop 1	192.0.0.5	255.255.255.0
PC 3	192.0.0.6	255.255.255.0

Step 5: For checking connectivity, select "Simple PDU packet" from right side of packet tracer & select source PC and ~~description~~ destination PC. You will get response at right bottom of the packet tracer window.

Fire	last status	Source	Destination	Type	Colour	Time	Period	N
⊙	Successful	PC 0	Laptop 0	ICMP	Yellow	0.00	N	0
⊙	Successful	PC 2	Laptop 1	ICMP	Green	0.00	N	1

Step 6: To get the ping response. Open desktop of host band select command prompt. Run the command as shown below.

```
pc > ping 192.0.0.1
pinging 192.0.0.1 with 32 bytes of data

Reply from 192.0.0.1 : bytes = 32 time=126ms TTL=128
Reply from 192.0.0.1 : bytes = 32 time = 64ms TTL=128
Reply from 192.0.0.1 : bytes = 32 time = 63ms TTL = 12
Reply from 192.0.0.1 : bytes = 32 time = 63ms TTL

ping statistics for 192.0.0.1
  packets: sent = 4, Received = 4, lost = 0 (0% loss)
Approximate round trip times in ms :-
  Minimum = 63ms, Maximum = 126 ms, Avg = 79ms
```

Staff 28/01/26